

# Chiral Models and Stochastic Quantization

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We study gauge theories coupled to massless Weyl fermions within the Stochastic Quantization approach. We extend this method showing how consistent Langevin equations can be written when chiral fermions are present. We also analyse how symmetries can be handled in this approach to quantization. Although the method we propose is intended to apply in all dimensions (2, 4, and others), we analyse for simplicity a two-dimensional example to explain how anomalies arise using the Stochastic Regularization scheme. We also discuss the solution of the Chiral Schwinger Model using this alternative quantization method which also leads to a consistent unitary theory for a certain range of an a priori undetermined regularization parameter. © 1988 Academic Press, Inc.

## 1. INTRODUCTION

In recent years the possibility of consistently quantizing gauge theories with anomalies [1] has been supported by different investigations. On the one hand the success of Jackiw and Rajaraman [2] in showing that a consistent, unitary, Lorentz invariant theory emerges from an anomalous two-dimensional gauge theory (the Chiral Schwinger Model) triggered many investigations of different models with chiral fermions [3]. On the other hand, Faddeev and Shatashvili [4] proposed a modification of canonical quantization of anomalous theories by adding a Wess–Zumino term to the gauge field and chiral fermion contributions, so that the total effective action would regain gauge invariance.

Soon after these important developments, it was observed that integration over all gauge field configurations in the path-integral approach to Quantum Field Theory makes the Wess–Zumino term naturally arise in chiral gauge models leading to anomaly free theories [5–7].

Although many subsequent investigations confirmed these results [8], there was first some confusion concerning the one parameter ambiguity [9] existing in the Chiral Schwinger Model, as well as in connection with the appearance of the Wess–Zumino term when integrating over all gauge fields [10]. Since most of the results were obtained within the path-integral approach to quantization, a confirmation using the canonical Hamiltonian formalism would be welcome. Although some results in this last framework have been reported [3, 11], the complete

reconstruction of consistent chiral theories from the Hamiltonian formalism is still lacking.

There is an alternative method for quantization developed after the proposal by Parisi and Wu [12] that considers Euclidean Quantum Field Theory as the equilibrium limit of a statistical system which evolves in a new fictitious time direction  $\tau$  until it reaches the equilibrium limit as  $\tau \rightarrow \infty$ . The method, known as Stochastic Quantization (SQM), has now grown into a useful tool in several areas of Quantum Field Theory [13]. In many cases, particularly those concerning gauge theories (no gauge fixing is in principle necessary), it has been shown to be a simple and elegant approach to quantization [14, 13].

It is then interesting to investigate chiral models using the SQM so as to obtain new insights into the problem of quantization of anomalous gauge theories. It is this question that we address in the present work. We first describe the method of Stochastic Quantization when Dirac fermions are present, following the approach of Refs. [15–19], putting particular emphasis on the issue symmetries within the SQM (Section 2).

In Section 3 we extend the SQM to the case of gauge theories with chiral fermions showing how Langevin equations governing the evolution in fictitious time must be modified when Weyl fermions are present. The method we propose is intended to apply in all dimensions (2, 4, and others); for simplicity we show how it works by evaluating the anomaly in the 2-dimensional Abelian case. Here again, the undetermined parameter introduced by Jackiw and Rajaraman [2] appears as a regularization ambiguity (in this case within the context of the Stochastic Regularization scheme [17–20]). We also discuss the exact solution of the Chiral Schwinger Model reobtaining via the SQM the results originally found by Jackiw and Rajaraman [2]: a consistent unitary theory emerges from an anomalous gauge theory provided the arbitrary parameter takes values in a certain range.

We then give a summary of our results in Section 4.

## 2. FERMIONS WITHIN THE STOCHASTIC QUANTIZATION SCHEME

We discuss in this section the method of Stochastic Quantization in the presence of fermion fields following the approach of Refs. [14–19]. After a brief review of the method in Section 2.1, we describe its application to a gauge theory with Dirac fermions (Section 2.2) putting special emphasis on the way symmetries are treated in this approach to quantization.

### 2.1 Review of Stochastic Quantization

In the SQM, as first proposed by Parisi and Wu [12] one considers a physical system in  $d$ -dimensional (Euclidean) space-time, with dynamics defined by an action  $S[\phi]$  of some (scalar, for simplicity) field  $\phi(x)$  and then one adds an extra dimension supplied by a fictitious time  $\tau$ ; the field  $\phi$  now takes values in a  $(d+1)$ -

dimensional space,  $\phi = \phi(x, \tau)$ . One then imposes an evolution in  $\tau$  governed by the Langevin equation

$$\frac{\partial \phi}{\partial \tau}(x, \tau) = -V[\phi] + \eta(x, \tau), \quad (2.1)$$

where  $\eta(x, \tau)$  is a gaussian noise and  $V[\phi]$  is a drift term related to  $S[\phi]$  which must be chosen so as to reach thermodynamic equilibrium as  $\tau \rightarrow \infty$ .

Due to Eq. (2.1),  $\phi$  becomes a stochastic field which we indicate by writing  $\phi_\eta(x, \tau)$ . Correlation functions of  $\phi_\eta$  are given by

$$\langle \phi_\eta(x_1, \tau_1) \phi_\eta(x_2, \tau_2) \cdots \phi_\eta(x_n, \tau_n) \rangle_\eta = \int \mathcal{D}\eta P[\eta] \prod_{i=1}^n \phi_\eta(x_i, \tau_i), \quad (2.2)$$

where  $P[\eta]$  is the probability distribution which can be written as

$$P[\eta] = \mathcal{N} \exp \left[ -\frac{1}{4} \int d^d x d^d y dt d\tau' \eta(x, \tau) K^{-1}(x, y) \eta(y, \tau') \right] \quad (2.3a)$$

$$\int \mathcal{D}\eta P[\eta] = 1 \quad (2.3b)$$

$$\langle \eta(x, \tau) \eta(x', \tau') \rangle_\eta = 2K(x, x') \delta(\tau - \tau') \quad (2.3c)$$

and  $K$  is a hermitian Kernel.

The connection with Quantum Field Theory is made by choosing  $V[\phi]$  in the form

$$V[\phi] = \int d^d y K(x, y) \frac{\partial S[\phi]}{\partial \phi(y, \tau)}. \quad (2.4)$$

Indeed, one can prove [12] that as  $\tau \rightarrow \infty$  the stochastic expectation value (2.2) coincides with the corresponding vacuum expectation value of the underlying Quantum Field Theory, i.e.,

$$\lim_{\tau \rightarrow \infty} \langle \phi_\eta(x_1, \tau_1) \cdots \phi_\eta(x_n, \tau_n) \rangle_\eta = \langle \phi(x_1) \cdots \phi(x_n) \rangle, \quad (2.5)$$

where the r.h.s. in (2.5) is given by

$$\langle \phi(x_1) \cdots \phi(x_n) \rangle = \frac{\int \mathcal{D}\phi e^{-S[\phi]} \prod_{i=1}^n \phi(x_i)}{\int \mathcal{D}\phi e^{-S[\phi]}}; \quad (2.6)$$

the choice of  $K$  can be made within a larger class of Kernels. It is interesting to note that concerning bosonic fields, a simple possible choice for  $K$  in (2.3), (2.4) is  $K(x, y) = \delta(x - y)$ . As explained below, when fermions are present a more

elaborated election must be done in order to ensure the existence of an equilibrium state for  $\tau \rightarrow \infty$ . Moreover, when a regularization is needed (for example, when performing a stochastic perturbation theory) the choice of an appropriate Kernel allows one to handle the problem of divergences respecting the relevant symmetries of the theory at all steps of the calculation.

Statement (2.5) can be easily proven [16] using the Fokker–Planck equation associated to the Langevin equation (2.1). In the former scheme, evolution in time is described by a probability distribution  $P[\phi, \tau]$  satisfying

$$-\frac{\partial P}{\partial \tau}(\phi, \tau) = H_{\text{FP}} P[\phi, \tau], \quad (2.7)$$

where the Fokker–Planck Hamiltonian is

$$H_{\text{FP}} = \int d^d x d^d y \frac{\delta}{\delta \phi(x)} K(x, y) \left[ \frac{\delta S}{\delta \phi(y)} + \frac{\delta}{\delta \phi(y)} \right]. \quad (2.8)$$

Equations (2.7) and (2.8) imply that

$$\lim_{\tau \rightarrow \infty} P[\phi, \tau] = \frac{e^{-S[\phi]}}{\int \mathcal{D}\phi e^{-S[\phi]}}, \quad (2.9)$$

which clearly shows the connection with Quantum Field Theory in the path-integral approach. In proving (2.9) it is important for  $H_{\text{FP}}$  to have a well-defined eigenvalue problem,  $H_{\text{FP}} \varphi_n = \lambda_n \varphi_n$  with  $\lambda_n \geq 0$  since  $P[\phi, \tau] \sim \sum_n a_n \varphi_n e^{-\lambda_n \tau}$ .

## 2.2. Gauge Theory with Dirac Fermions

We shall now describe how the SQM presented above can be extended to include Dirac fermions. We shall place special emphasis on those aspects related to the way symmetries must be handled within this framework. Although these aspects have been already treated by several authors [19–25, 13] we think it is worthwhile to discuss them in detail, particularly in view of the treatment of Weyl fermions that we present in the next section.

Let us consider a gauge theory with (Euclidean)  $d$ -dimensional action  $S$  given by

$$S = \int d^d x \left( -\frac{1}{4} F_{\mu\nu}^2 + \bar{\psi} \not{D}[A] \psi \right) \quad (2.10)$$

$$\not{D}[A] = i \not{\partial} + e \not{A}$$

with the gauge field  $A_\mu$  taking values in the Lie Algebra of some group  $G$  and the massless fermions  $\psi$  in the fundamental representation of  $G$ .

The SQM described in (2.1) can be generalized when Dirac fermions are included as follows. The fields  $\psi$  and  $\bar{\psi}$  are treated as independent Grassmann variables and the Langevin equations read

$$\frac{\partial \psi(x, \tau)}{\partial \tau} = - \int d^d y K(x, y) \frac{\delta S}{\delta \bar{\psi}(y, \tau)} + \Theta(x, \tau) \quad (2.11a)$$

$$\frac{\partial \bar{\psi}(x, \tau)}{\partial \tau} = - \int d^d y \frac{\delta S}{\delta \psi(y, \tau)} \bar{K}(x, y) + \bar{\Theta}(x, \tau) \quad (2.11b)$$

$$\frac{\partial A_\mu(x, \tau)}{\partial \tau} = - \frac{\delta S}{\delta A_\mu(x, \tau)} + \eta_\mu(x, \tau). \quad (2.12)$$

Functions  $\Theta$  and  $\bar{\Theta}$  are now anticommuting spin  $\frac{1}{2}$  noise fields. Note that for the gauge field equation (2.12), we have made the simplest choice of Kernel,  $K(x, y) = \delta(x - y)$ . A similar choice for the fermionic Kernel in (2.11) is not possible since, fermions being massless, the corresponding Langevin equation will not generate the proper equilibrium configuration as  $\tau \rightarrow \infty$  [15–18]. Moreover, when the Stochastic Regularization scheme is adopted in order to handle divergences, the choice of the Kernel is crucial to ensuring gauge-invariance at all steps of the calculations.

It is important to stress that the behavior of the noise fields in (2.11) under symmetry transformations leaving invariant  $S$  (in particular gauge-transformations) is not a priori determined. However, once Kernel  $K$  is selected, the variations of physical fields  $\psi, \bar{\psi}$ , and  $A_\mu$  fix the way  $\Theta, \bar{\Theta}$  transform. When stochastic regularization is adopted it is necessary to select a  $K$  respecting gauge-covariance. The natural choice in this case is

$$\begin{aligned} K(x, y; A) &= \delta(x - y) \not{D}[A] \\ \bar{K}(x, y; A) &= \not{\bar{D}}[A] \delta(x - y). \end{aligned} \quad (2.13)$$

Of course, gauge-invariance will be ensured by a general Kernel  $K$  transforming covariantly under gauge transformations

$$K(x, y; A') = e^{i\delta\alpha} K(x, y; A) e^{-i\delta\alpha}, \quad (2.14)$$

where  $\exp(i\delta\alpha)$  is an infinitesimal gauge-transformation and

$$A'_\mu = A_\mu + \frac{i}{e} D_\mu \delta\alpha. \quad (2.15)$$

Once a choice of the kind in (2.14) is done, Langevin equations (2.11) determine the transformation law for  $\Theta$  and  $\bar{\Theta}$  under gauge-rotations. Indeed, transforming the solutions  $\psi_\Theta, \bar{\psi}_\Theta$  of Eqs. (2.11),

$$\begin{aligned} \psi_\Theta &\rightarrow \psi'_\Theta = e^{i\delta\alpha} \psi_\Theta = \psi_\Theta + i \delta\alpha \psi_\Theta \\ \bar{\psi}_\Theta &\rightarrow \bar{\psi}'_\Theta = \bar{\psi}_\Theta e^{-i\delta\alpha} = \bar{\psi}_\Theta - i \bar{\psi}_\Theta \delta\alpha, \end{aligned} \quad (2.16)$$

one obtains new fields  $\psi'_\Theta, \bar{\psi}'_\Theta$  which must be solutions of Langevin equations with transformed noise functions  $\Theta', \bar{\Theta}'$ ,

$$\frac{\partial \psi'_\Theta}{\partial \tau} = - \int d^d y K(x, y; A') \frac{\delta S}{\delta \bar{\psi}'} + \Theta', \quad (2.17)$$

and a similar equation for  $\bar{\psi}'_\Theta$ . Since  $K$  transforms covariantly, we have

$$\int d^d y K(x, y; A) \frac{\delta S}{\delta \bar{\psi}'} = e^{i\delta\alpha(x)} \int d^d y K(x, y; A) \frac{\delta S}{\delta \bar{\psi}} \quad (2.18)$$

and hence

$$\frac{\partial \psi_\Theta}{\partial \tau}(x, \tau) = - \int d^d y K(x, y; A) \frac{\delta S}{\delta \bar{\psi}(y, \tau)} + e^{-i\delta\alpha} \Theta'(x, \tau). \quad (2.19)$$

Subtracting (2.11a) from (2.19) we have

$$\Theta'(x, \tau) = e^{i\delta\alpha(x)} \Theta(x, \tau) \quad (2.20a)$$

and analogously

$$\bar{\Theta}'(x, \tau) = \bar{\Theta}(x, \tau) e^{-i\delta\alpha(x)}. \quad (2.20b)$$

We then see that the noise functions  $\Theta$  and  $\bar{\Theta}$  transform like the fermionic fields  $\psi$  and  $\bar{\psi}$ . A similar reasoning applies for any symmetry transformation leaving invariant the action  $S$ . In particular, concerning  $U(1)$  global chiral transformations,

$$\psi'_\Theta = e^{\gamma_5 \delta\beta} \psi_\Theta, \quad \bar{\psi}'_\Theta = \bar{\psi}_\Theta e^{\gamma_5 \delta\beta}, \quad (2.21)$$

one has

$$\Theta' = e^{\gamma_5 \delta\beta} \Theta, \quad \bar{\Theta}' = \bar{\Theta} e^{\gamma_5 \delta\beta} \quad (2.22)$$

provided

$$K(x, y; A) e^{\gamma_5 \delta\beta} = \bar{e}^{\gamma_5 \delta\beta} K(x, y; A). \quad (2.23)$$

Note that in many investigations [20, 21] these transformation laws were assumed in order to obtain the correct anomaly. With the formalism presented in this section quantization of fermionic systems can be performed in a simple and elegant way; in particular, the question of symmetries can be analyzed in detail. We discuss this question as well as the appearance of anomalies in the next section.

### 3. CHIRAL THEORIES

Gauge theories coupled to massless (Weyl) fermions are afflicted by anomalies in space-times with dimensionality  $d = 2 + 4k$ ,  $k = 0, 1, \dots$ . In particular, the Noether

current associated with chiral symmetry ceases to be conserved after quantization. This problem is also reflected in the gauge non-invariance of the effective action obtained by integration over fermions in the path-integral approach. Since anomalies were thought to spoil consistency and unitarity (and renormalizability), anomaly cancellation mechanisms had to be developed to attain a consistent quantum theory [1].

Recently, it has been shown that chiral gauge theories may be consistently quantized [2–7] using the path-integral approach, or in some cases the canonical Hamiltonian formalism. Although complete success has been attained in 2-dimensional models, many open problems remain, particularly concerning renormalizability of 4-dimensional theories.

We present in this section an extension of the SQM intended to apply to gauge theories coupled to Weyl fermions. We first show how the Langevin equations must be handled when chiral fermions are present and illustrate our proposal by evaluating the anomaly in the Schwinger Chiral Model. Then, we solve this last model and reobtain the Jackiw–Rajaraman [2] result concerning its consistency at the quantum level, within the SQM.

### 3.1. Setting the Langevin Equations

We now consider a gauge theory coupled to Weyl fermions (for definiteness left-handed ones):

$$\psi = -\gamma_5 \psi. \quad (3.1)$$

In this case, the fermionic action  $S_F$  can be written in the form

$$S_F = \int d^d x \bar{\psi} D[A] \psi \quad (3.2)$$

with

$$D[A] = (i\not{D} + e\not{A}) \frac{(1 - \gamma_5)}{2}. \quad (3.3)$$

Note that the Dirac operator  $D[A]$  maps negative chirality spinors into positive chirality ones and consequently it does not have a well-defined eigenvalue problem.

In the path-integral approach to quantization this poses a problem since the fermionic generating function is defined as the Dirac operator determinant (i.e., a product of eigenvalues). To handle this difficulty, one usually modifies the model by adding free right-handed fermions in order to work with a new Dirac operator, having a well-defined eigenvalue problem.

In the present approach this problem can be overcome without modifying the Dirac operator  $D[A]$ , by an appropriate choice of Kernel  $K$ . Indeed if one chooses a Kernel  $K$  mapping positive chirality fermions into negative chirality ones, the composition of  $K$  with  $D[A]$  in Langevin eqs. (2.11) ensures that one remains in the left-handed sector. This avoids doubling in the number of degrees of freedom

implied by addition of right-handed fermions. In this respect, our proposal is much the same as that recently presented by Singer [27] to handle the problem of chiral determinants, by composing the Dirac operators applied to left-handed fermions with a fixed one applied to right-handed fermions so as to have a well-defined eigenvalue problem.

The point is now to find an adequate Kernel  $K$  in order to have consistent Langevin equations. Obviously  $D[A]$  is no longer a possible candidate, as it was in the Dirac fermion case ( $KD = D^2 = 0$  and hence Langevin eqs. (2.11) become trivial).

We then propose, instead of (2.13),

$$\begin{aligned} K(x, y; A) &= \delta(x - y) \tilde{\mathcal{D}}[A] \left( \frac{1 + \gamma_5}{2} \right) \\ \bar{K}(x, y; A) &= \tilde{\mathcal{D}}[A] \left( \frac{1 + \gamma_5}{2} \right) \delta(x - y), \end{aligned} \quad (3.4)$$

where  $\tilde{\mathcal{D}}[A] = \gamma_\mu \tilde{D}_\mu[A]$  must be chosen with only the restriction of ensuring convergence as  $\tau \rightarrow \infty$ .

To illustrate how anomalies arise in this framework we now discuss a simple two-dimensional example: the Chiral Schwinger Model [2].

### 3.2. The Anomaly in the Chiral Schwinger Model

Consider two-dimensional quantum electrodynamics with Weyl fermions. The dynamics of this model is defined by the action

$$S = -\frac{1}{4} \int d^2x F_{\mu\nu}^2 + S_F, \quad (3.5)$$

where the fermionic action  $S_F$  is given by Eq. (3.2) with  $d = 2$ . Our conventions for the  $\gamma$ -matrices are

$$\begin{aligned} \gamma_0 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & \gamma_1 &= \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix} \\ \gamma_5 &= i\gamma_0\gamma_1 \end{aligned} \quad (3.6)$$

and the following relations hold,

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \quad \gamma_\mu \gamma_5 = i\epsilon_{\mu\nu} \gamma_\nu. \quad (3.7)$$

We now must choose an adequate Kernel as in (3.4). Inspired in the path-integral [3–8] approach to two-dimensional chiral models, we propose

$$K(x, y; A) = \delta(x - y) (i\not{D} + e\alpha\not{A}) \left( \frac{1 + \gamma_5}{2} \right), \quad (3.8)$$



where  $\alpha$  is an a priori free parameter which is related, as we shall see, with the one introduced by Jackiw and Rajaraman [2]. The Langevin equations then read

$$\frac{\partial \psi_{\Theta}}{\partial \tau} = -(i\cancel{D} + e\alpha A)(i\cancel{D} + eA) \left( \frac{1 - \gamma_5}{2} \right) \psi_{\Theta} + \Theta \quad (3.9)$$

$$\frac{\partial \bar{\psi}_{\Theta}}{\partial \tau} = -\bar{\psi}_{\Theta} \left( \frac{1 + \gamma_5}{2} \right) (i\cancel{D} + eA)(i\cancel{D} + e\alpha A) + \bar{\Theta}$$

$$\frac{\partial A_{\mu}}{\partial \tau} = \partial_{\nu} F_{\nu\mu} + e\bar{\psi}_{\Theta} \left( \frac{1 + \gamma_5}{2} \right) \gamma_{\mu} \psi'_{\Theta} + \eta_{\mu}. \quad (3.10)$$

In order to evaluate the anomaly we consider the following infinitesimal gauge-transformation:

$$\begin{aligned} \psi_{\Theta} &= e^{ie(x)} \psi'_{\Theta} \\ \bar{\psi}_{\Theta} &= \bar{\psi}'_{\Theta} e^{-ie(x)}. \end{aligned} \quad (3.11)$$

Under these transformations, the fermionic action changes by

$$\delta S_F = \int d^2x \, \varepsilon(x) \, \partial_{\mu} J_{\mu}(x) \quad (3.12)$$

with

$$J_{\mu} = \frac{1}{2} \bar{\psi} \gamma_{\mu} (1 - \gamma_5) \psi. \quad (3.13)$$

Since a regularization of divergences will be necessary we adopt the so-called Stochastic Regularization method [10–20] which consists of replacing the stochastic averages (2.3c) by

$$\langle \Theta(x, \tau) \bar{\Theta}(x', \tau') \rangle_A = 2\tilde{\cancel{D}}[A] \frac{(1 + \gamma_5)}{2} a_A(\tau - \tau') \quad (3.14)$$

with  $a_A$  satisfying

$$\begin{aligned} a_A(\tau) &= a_A(-\tau), \quad \int d\tau' a_A(\tau - \tau') = 1 \\ \lim_{A \rightarrow \infty} a_A(\tau - \tau') &= \delta(\tau - \tau'). \end{aligned} \quad (3.15)$$

The vacuum expectation value of (3.12) is given by the SQM as

$$\langle \delta S \rangle = \lim_{\tau \rightarrow \infty} \langle \delta S_{\Theta}[\psi_{\Theta}(x, \tau), \bar{\psi}_{\Theta}(x, \tau)] \rangle_{\Theta}. \quad (3.16)$$

Now,  $\delta S_{\Theta}$  can be written in the form

$$\delta S_{\Theta} = -\frac{1}{2} \int d^2x \, \bar{\psi}_{\Theta} \left( \frac{1 + \gamma_5}{2} \right) [D[A], \varepsilon(x)] \psi_{\Theta}. \quad (3.17)$$

Introducing the solutions  $\psi_\theta(x, \tau)$ ,  $\bar{\psi}_\theta(x, \tau)$  of Langevin equations (3.9),

$$\begin{aligned}\psi_\theta(x, \tau) &= \int_0^\tau d\tau' \exp \left[ -(i\bar{\not{D}} + e\alpha A)(i\bar{\not{D}} + eA) \left( \frac{1-\gamma_5}{2} \right) (\tau - \tau') \right] \Theta(x, \tau') \\ \bar{\psi}_\theta(x, \tau) &= \int_0^\tau d\tau' \bar{\Theta}(x, \tau') \exp \left[ - \left( \frac{1+\gamma_5}{2} \right) (i\bar{\not{D}} + eA)(i\bar{\not{D}} + e\alpha A) (\tau - \tau') \right],\end{aligned}\quad (3.18)$$

in (3.16) one gets after some algebra

$$\begin{aligned}\langle \delta S \rangle_\theta &= -\frac{1}{2} \int d^2x d^2y \delta(x-y) \int_0^\tau d\tau_1 d\tau_2 \\ &\times \text{tr} \{ e^{-(D\bar{D}_x)(\tau-\tau_1)} (1+\gamma_5) [\not{D}_x, \varepsilon(x)] e^{-(D_x\bar{D})(\tau-\tau_2)} \langle \Theta(x, \tau_2) \bar{\Theta}(y, \tau_1) \rangle_\theta \},\end{aligned}\quad (3.19)$$

where

$$\begin{aligned}\not{D} &= (i\bar{\not{D}} + eA) \\ D_\alpha &= (i\bar{\not{D}} + e\alpha A) \left( \frac{1+\gamma_5}{2} \right).\end{aligned}\quad (3.20)$$

Using Eq. (3.14) for  $\langle \Theta, \bar{\Theta} \rangle_A$ , one finds

$$\begin{aligned}\langle \delta S \rangle_{\theta, A} &= - \int d^2x d^2y \frac{d^2k}{(2\pi)^2} \int_0^\tau d\tau_1 d\tau_2 \delta(x-y) a_A(\tau_1 - \tau_2) \varepsilon(x) \\ &\times \text{tr} \{ (1-\gamma_5) D_\alpha \not{D} e^{-D_x\bar{D}(2\tau-\tau_1-\tau_2)} \\ &- (1+\gamma_5) \not{D} D_\alpha e^{-\bar{D}D_x(2\tau-\tau_1-\tau_2)} \} e^{ik(x-y)},\end{aligned}\quad (3.21)$$

where we indicate with the subscript  $A$  in the l.h.s. that the limit  $A \rightarrow \infty$  has not been done yet. In fact, in the Stochastic Regularization scheme, one first performs the  $\tau \rightarrow \infty$  limit and then the  $A \rightarrow \infty$  one. It is at this point that the Kernel  $K$  plays a central role: on one hand, it must ensure convergence of exponentials in (3.21) when  $\tau \rightarrow \infty$ . This is precisely what happens with our choice (3.8):

$$\begin{aligned}\lim_{\tau \rightarrow \infty} \langle \delta S \rangle_{\theta, A} &= i \int d^2x d^2k \frac{\varepsilon(x)}{2} \\ &\times \text{tr} \{ (1-\gamma_5) e^{-d_\alpha(k)d(k)/A^2} - (1+\gamma_5) e^{-d(k)d_\alpha(k)/A^2} \},\end{aligned}\quad (3.22)$$

where

$$\begin{aligned}d(k) &= \not{D} - \not{k} \\ d_\alpha(k) &= D_\alpha - \not{k}.\end{aligned}\quad (3.23)$$

We have used properties (3.15) to make the  $\Lambda$ -regulator appear explicitly. At this point, it is interesting to note the similarity between expression (3.22) obtained in Stochastic Regularization and the corresponding one evaluated using the heat-kernel approach. As in the latter scheme it is the principal symbol of  $d(k)$   $d_\alpha(k)$  and  $d_\alpha(k)$   $d(k)$  which determines the convergences of (3.22) when  $\Lambda \rightarrow \infty$  [28, 29]. Rescaling variables ( $k \rightarrow k/\Lambda$ ) and taking this limit we finally get

$$\langle \delta S \rangle = \frac{e}{4\pi} \int d^2x \varepsilon(x) [(1 - \alpha) \partial_\mu A_\mu + (1 + \alpha) i \varepsilon_{\mu\nu} F_{\mu\nu}]. \quad (3.24)$$

Comparing this with (3.12), we get for the anomaly

$$\partial_\mu J_\mu = \frac{e}{4\pi} [(1 - \alpha) \partial_\mu A_\mu + (1 + \alpha) i \varepsilon_{\mu\nu} F_{\mu\nu}]. \quad (3.25)$$

As first explained in Ref. [2] no value of  $\alpha$  gives a gauge-invariant answer.

We can now show how the Jackiw and Rajaraman results on the consistency of the Chiral Schwinger Model can be reobtained in the context of Stochastic Quantization. Following Webb's analysis for the ordinary Schwinger Model [25] one incorporates the anomaly result (3.25) in the Langevin equation (3.10), where the current appears explicitly; this gives an effective drift term for the evolution of  $A_\mu(x, \tau)$ . In fact, the Langevin equation (3.10)

$$\frac{\partial A_\mu}{\partial \tau} = \partial_\nu F_{\nu\mu} + J_\mu + \eta_\mu \quad (3.26)$$

can be rewritten in terms of two scalar fields  $\phi$  and  $\rho$  through

$$A_\mu(x, \tau) = \varepsilon_{\mu\nu} \partial_\nu \phi(x, \tau) + \partial_\mu \rho(x, \tau). \quad (3.27)$$

Multiplying (3.26) by  $\varepsilon_{\mu\sigma} \partial_\sigma$  and by  $\partial_\mu$  and using (3.25) for the anomaly, one can identify the equations of motion for  $A_\mu$  from the resulting drift term:

$$\begin{aligned} \square \phi + g(1 - \alpha) \square \phi + ig(1 + \alpha) \square \rho &= 0 \\ g(1 - \alpha) \square \rho + ig(1 + \alpha) \square \phi &= 0 \end{aligned} \quad (3.28)$$

$$g = \frac{e^2}{4\pi}.$$

Diagonalizing (3.28) one concludes that the theory corresponds to that of a free scalar particle  $\phi$  (with mass  $m^2 = (e^2/2\pi)(1 + \alpha^2)/(\alpha - 1)$ ) provided  $\alpha > 1$  plus a massless excitation. This coincides with the results of Ref. [2] concerning the fact that the theory is consistent and unitary, provided an undetermined parameter (which appears due to regularization ambiguities) takes values in a certain range ( $\alpha > 1$  in the present case, so  $m$  does not correspond to a tachyonic mass).

## 4. SUMMARY AND CONCLUSIONS

We have discussed in this work quantization of gauge theories with Weyl fermions using the Stochastic Quantization Method, motivated by the recent interest [1] in quantization of anomalous gauge theories.

We have extended the SQM for the case in which Weyl fermions are present, showing that an adequate choice of the a priori arbitrary Kernel  $K$  appearing in Langevin Eqs. (2.11) solves the problem posed by the fact the Dirac operator maps negative chirality fermions into positive chirality ones. Indeed, election (3.4) for  $K$  leads to consistent Langevin equations (i.e., with all terms having the same chirality). As we stated above, this treatment of chiral fermions resembles the one proposed by Singer [27].

Although the proposed method is intended to apply in all dimensions (2, 4, and others) for simplicity we have applied it to the two-dimensional case showing how the anomaly arises in this context.

Finally, the results for the solution of the Chiral Schwinger Model are a confirmation, using an alternative quantization procedure, of the emergence of a consistent and unitary theory from an anomalous gauge model as advocated in Ref. [2]. Of course, the question of renormalizability of realistic  $d=4$  models remains to be investigated. We expect that the present method can help in giving an answer.

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